

What it takes to win

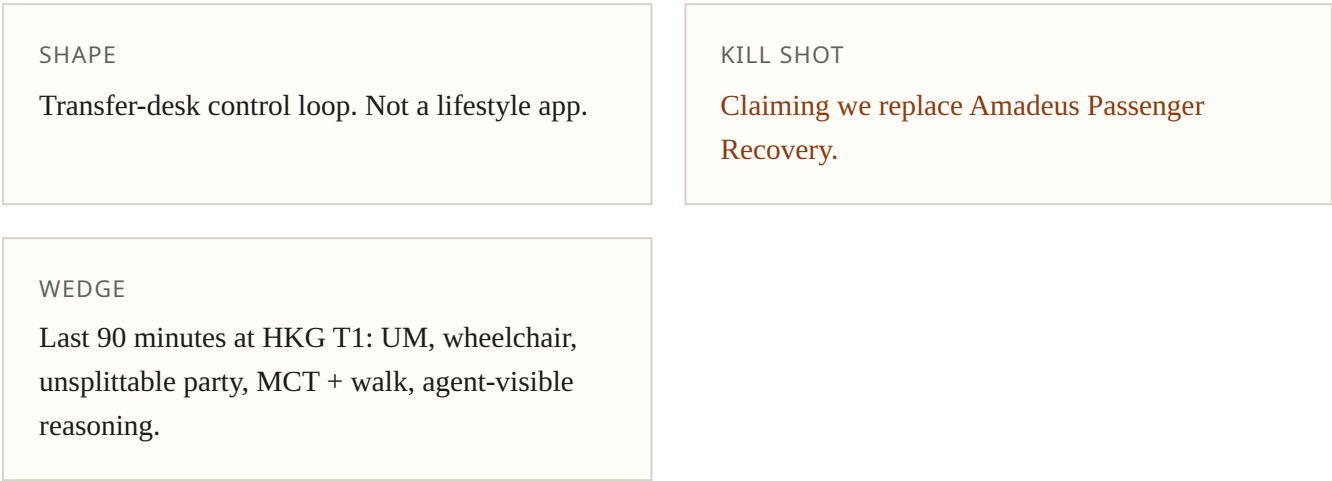
A briefing for teammates who already know how to build. Judging this year is feasibility and Business Units. The kill shot is claiming we replace Passenger Recovery. The wedge is the last 90 minutes at HKG T1.

Start here (one minute)

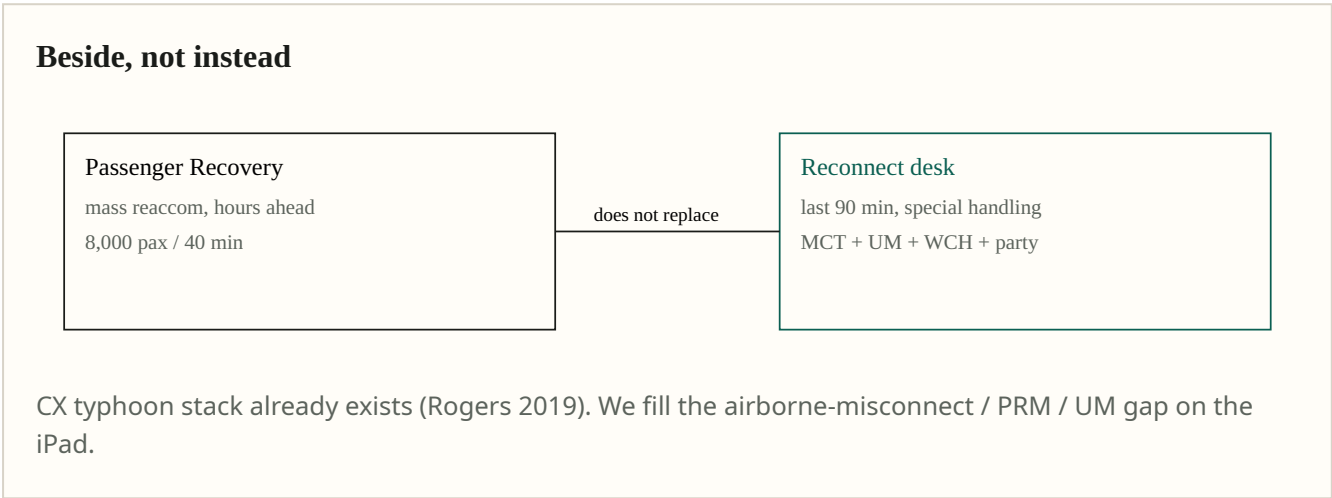
2026 judges score **feasibility**, **Business Units**, **path to adoption**. Apply 2026-09-27; 24-hour 2026-11-14/15; pitch 2026-11-15; this repo’s demo 2026-11-16.

Reconnect is a **desk tool**, not a lifestyle app. FlyLab (employee, existing ops, humans decide) is the analogue. U-Explore is not.

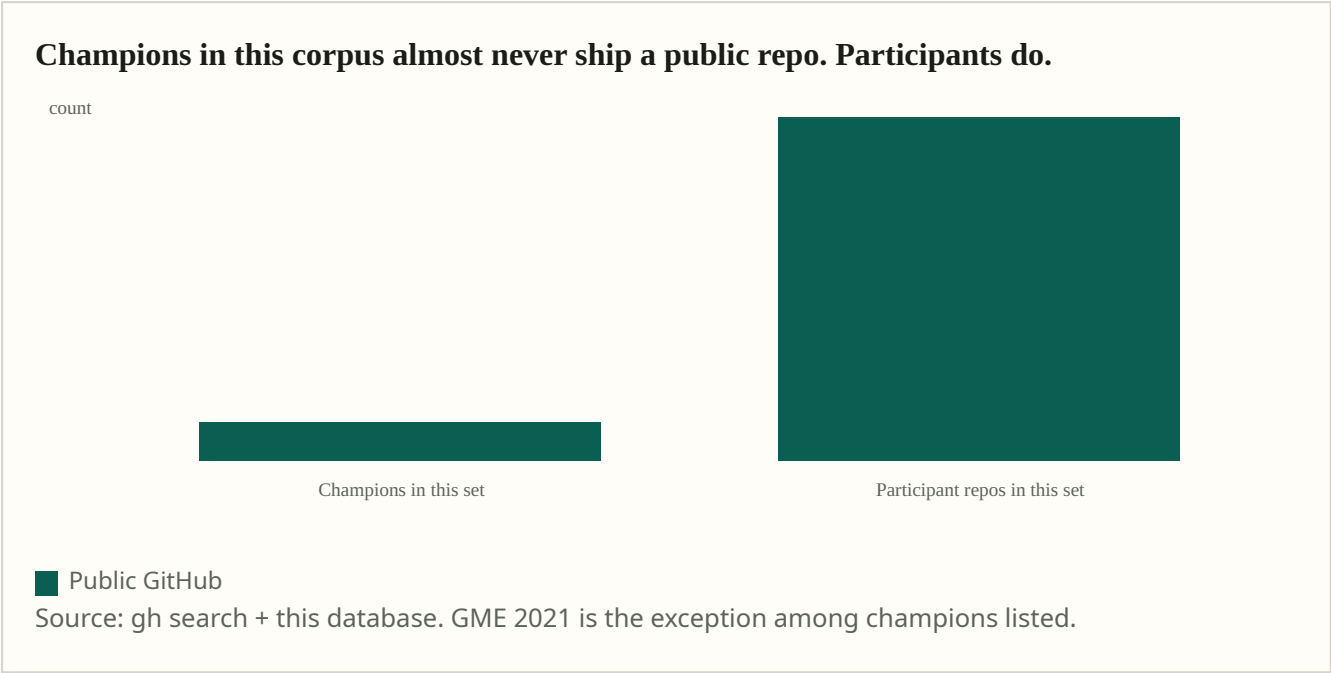
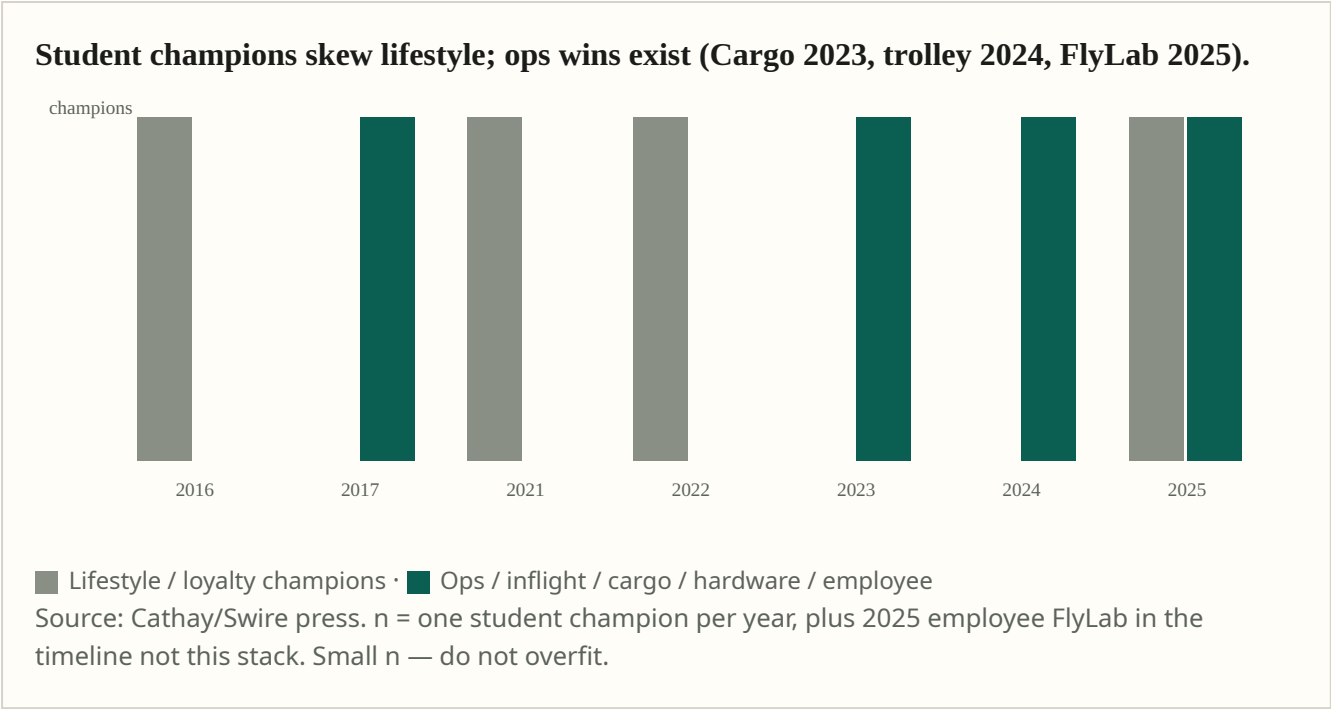
Incumbent objection: CX already bought mass IROPS (Passenger Recovery + Accenture rebooking bots + 15below). Pitch **beside**, not instead. The wedge is the last 90 minutes at HKG T1.



2018 overall champion name was not found in Cathay press. Anita and Lugless are sourced as runners-up only.



What the record actually shows



Patterns that paid

Named BU beats generic AI

WINNERS Attach to HK Express, Cargo, Inflight, or Flight Ops in the first sentence.

WE HAVE HKG transfer desk / Airport Systems is the owner. Say that, not “AI for aviation”.

SATURDAY Open with the desk, the queue, and Q77 — not the model.

90-second theater

WINNERS Judges must see the product: trolley, robot, briefing card, typhoon button.

WE HAVE Simulate Typhoon Delay and Late Inbound CX254 already refresh the live queue.

SATURDAY Narrate the six-step demo as ops, not as a tech tour.

Dual beneficiary

WINNERS Passenger care plus a second ledger: revenue, labour, ESG data, or safety.

WE HAVE Passenger protection plus desk throughput in a typhoon bank.

SATURDAY Name both: misconnect care and minutes off the transfer queue.

Feasibility 2026 is not lifestyle 2025

WINNERS BU judges now score adoption path. FlyLab (pilots + IT, summarise existing briefings) is the closer analogue than U-Explore.

WE HAVE Deterministic engine, Altéa/FIDS/CANS pilot note, no Altéa write in week 1.

SATURDAY We sit beside Passenger Recovery. We do not replace it.

Champions rarely open-source

WINNERS Win on pain, BU, and theater. Public GitHub is mostly participant trip planners.

WE HAVE A real repo with tests. Do not treat GitHub stars as the scoreboard.

SATURDAY Show the iPad and the reasoning strings, not a star count.

Delay re-plan without a desk owner loses

WINNERS Fight4Flight (2017 runner-up) notified and re-planned from the passenger side. It did not win.

WE HAVE Agent-in-the-loop on an iPad. Approve is the only send.

SATURDAY The missing 2017 piece was the transfer desk. That is the product.

Cathay already bought mass IROPS

WINNERS Do not pitch into a purchased capability as if the market were empty.

WE HAVE Last-90-minute special handling: UM, wheelchair, unsplittable party, MCT + walk.

SATURDAY Passenger Recovery does 8,000 pax / 40 min. We do tight connections the mass tool cannot walk-time.

LLM-decides-rebooking is a foil

WINNERS BU and IT will not let a model pick the flight.

WE HAVE Engine decides. LLM drafts. Flight-number guard. reasoning[] is verbatim.

SATURDAY The guard is the product, not ChatGPT.

Control cycle, station clock, sim, kill switch

WINNERS SpaceX-class ops: read inputs, compute state, write outputs, wait for the clock. Config for non-software engineers.

WE HAVE FIDS + PNR in, slack/MCT state, options + draft out. MCT table is the DSL. Sim seed.

RECONNECT_PILOT=off.

SATURDAY Say it in ops English, not brand English.

Walk time and special handling are the wedge

WINNERS ConnectionSaver treated gate walk as a product (SMS with minutes). MCT tables lie for PRM and UM.

WE HAVE Gate walk +15 WCH +20 UM, CX-metal-only for UM, unsplittable party, downgrade protection.

SATURDAY Mass reaccom does not escort an unaccompanied minor through HKG T1.

Cathay winners 2016–2025

2016

champion
no public repo

cxDiscovery · Inflight / Lifestyle
Personalised inflight e-magazine plus vendor targeting.
State a second beneficiary (desk throughput), not only passenger care.

2017

runner-up
no public repo

Aviator · Airport / connections
Dining and activity manager aimed at connecting passengers.
Connecting passengers are the moment; recovery is the job.

runner-up
no public repo

Fight4Flight · Customer experience
Delay notification and re-planning from the passenger side.
Passenger-only re-plan loses. The desk is the missing 2017 piece.

champion
no public repo

TouchCX · Inflight
Passenger phone to crew tablet for amenity requests (NFC).
Win by sitting on a crew/agent device, not another passenger super-app.

2018

runner-up
no public repo

Anita chatbot · Digital / CX
Chatbot that reminds passengers before boarding.
Notify is not recover. We notify after an agent approves a protected flight.

runner-up
no public repo

Lugless · Baggage / CX
Bags packed and waiting at destination.
Bags are out of SCOPE. Do not add them before Saturday.

2021

champion

GME / Asia Miles visualiser · Loyalty
Map and proximity for Asia Miles partners, plus redemption notices.
A visual of a real Cathay ledger (queue, slack) beats a generic dashboard.

2022

champion
no public repo

Cathay Green · Sustainability / Lifestyle
Currency and AI to promote carbon offset and green practice.
Only claim sustainability if we can count fewer stranded hotel nights. Do not fake it.

2023

champion
no public repo

NAAR cargo dimensioning · Cargo
Computer vision to size cargo and pack denser, with less manual labour.
Lead with a number (minutes, pax, queue length). NAAR led with accuracy and labour.

2024

champion
no public repo

GingTrip modular trolley · HK Express / Inflight
Modular galley trolley that sorts cabin waste and feeds a dashboard.
Do not add a robot. Do make the typhoon click as tactile as a trolley.

2025

champion
no public repo

FlyLab pre-flight briefings · Flight Ops
AI summaries of existing pre-flight briefing packs for pilots.
Closest 2025 analogue. Pair with a transfer agent. Do not invent a new PSS.

runner-up
no public repo

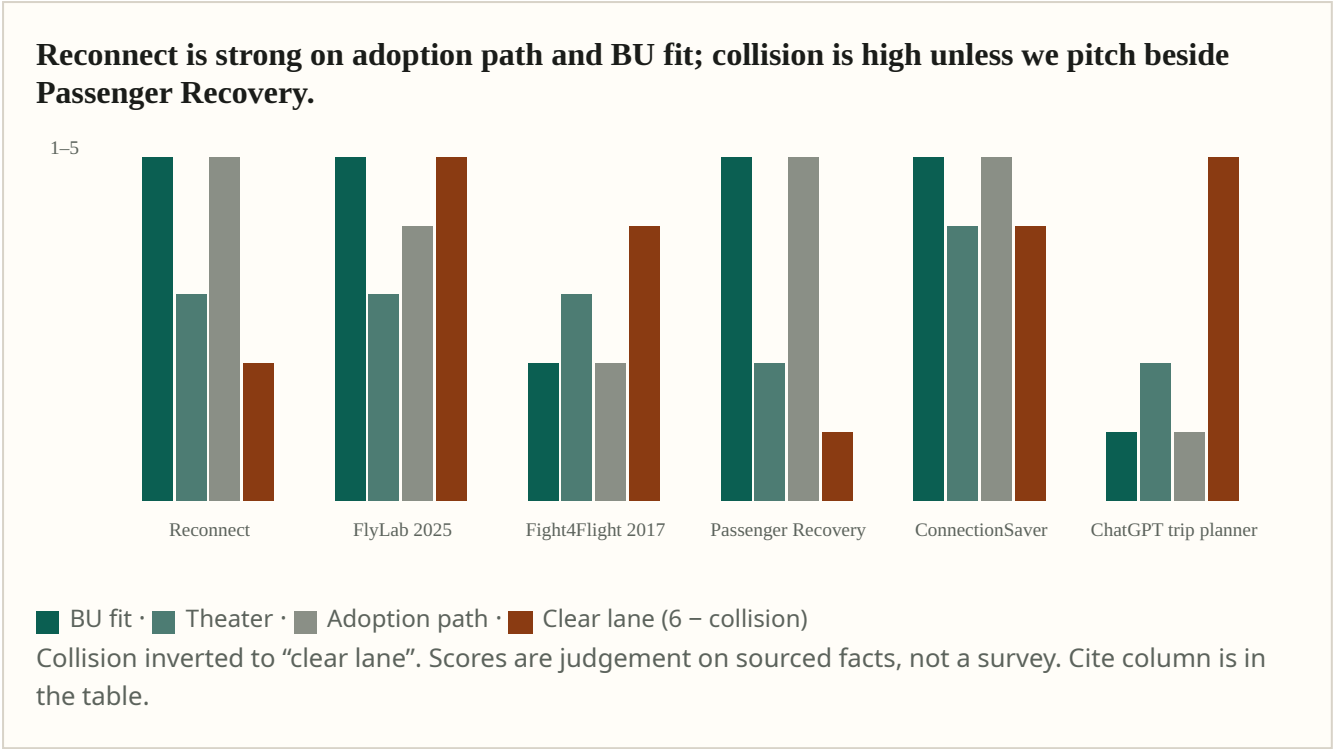
Niki Jr. · Customer experience / heritage
Polaroid robot named after CX DC-3 Niki; app booking, crowd-aware queueing.
Presentation award is theater. We already have buttons; narrate them.

champion
no public repo

U-Explore (Kairos Takeoff Crew) · HK Express
All-in-one HK Express customer-journey platform: loyalty, community, gamified content.
Do not copy the loyalty app. Copy the named BU and the lived pain sentence.

Reconnect scorecard

Engine citations: packages/engine/src/mct.ts (CX_CX 50 min, walk 10, WCH 15, UM 20), score.ts (tier×3 + seat×2 – delay/10), apps/console/lib/llm/guard.ts, docs/PILOT_PROPOSAL.md.



	BU	Theater	Path	Collision	Cite
Reconnect	5	3	5	4	mct.ts CX_CX 50 + walk 10; WCH +15; UM +20; score.ts; guard.ts; PILOT_PROPOSAL.md
FlyLab 2025	5	3	4	1	Employee track: pilots + IT, summarise existing briefings.
Fight4Flight 2017	2	3	2	2	Passenger re-plan, no desk. Runner-up.
Passenger Recovery	5	2	5	5	8,000 pax / 40 min. CX production ~2018.
ConnectionSaver	5	4	5	2	Walk-time SMS. Hold, not rebook.
ChatGPT trip planner	1	2	1	1	angusf777 CathayConnect. Foil.

1-5. Collision is inverted in the chart (clear lane = 6 – collision).

How we win — tagged

PITCH	L01. Open with Airport Systems / HKG transfer desk, not “AI for aviation”.
PITCH	L02. Name two beneficiaries: the passenger and the typhoon queue.
DEMO	L03. Narrate demo steps 3–6 as ops theater (typhoon, CX254, reasoning, approve).
LATER	L04. Do not add hardware or a robot. FlyLab won without one.
PITCH	L05. Use coach language: Q77 + Altéa cryptic is the bottleneck.
PITCH	L06. Claim ESG only if we can count fewer stranded nights. Otherwise omit.
PITCH	L07. Agent-in-the-loop is why Fight4Flight lost and we should not.
PITCH	L08. Pair the pitch with a real CX role (transfer agent + IT), FlyLab-style.
DEMO	L09. Say: LLM drafts; engine decides; guard strips hallucinated flights.
PILOT	L10. Sit beside Passenger Recovery as the last-90-minute special-handling layer.
PILOT	L11. Pilot metrics already in PILOT_PROPOSAL: 5s queue, reasoning not from LLM, flight-number-safe push.
PITCH	L12. Call the engine a control cycle on station time. MCT table is the DSL.
PITCH	L13. Seeded sim = HOOTL. Live adapter = HITL. RECONNECT_PILOT=off is abort.
DEMO	L14. Describe the iPad as one glass (queue, connection, action). Open MCT analogue.
PITCH	L15. Walk + WCH + UM is the wedge. ConnectionSaver made walk-time a product.
LATER	L16. Hold-the-outbound is a different decision. Park it in LATER. Know it exists.
LATER	L17. Do not import MILP aircraft-crew recovery. That is AOCC, not T1.
PILOT	L18. Do not scrape public FIDS. Pilot uses operational MQTT. OpenSky is research-only.
PITCH	L19. Use ChatGPT trip planners and GPT compensation as foils, not as features.
PITCH	L20. Do not invent impact numbers. NAAR and ConnectionSaver led with real ones; we do not have them yet.

Project explorer

cxDiscovery

2016 · cathay-winner · champion · copy-pitch · high

Personalised inflight e-magazine plus vendor targeting.

Strength. Named inflight publications as the pain

Weakness. Not an ops tool

Takeaway. State a second beneficiary (desk throughput), not only passenger care.

No public repo

[Cathay inaugural hackathon 2016](#)

TouchCX

2017 · cathay-winner · champion · copy-pitch · high

Passenger phone to crew tablet for amenity requests (NFC).

Strength. Crew workflow, not a concierge chatbot

Weakness. Cabin not transfer desk

Takeaway. Win by sitting on a crew/agent device, not another passenger super-app.

No public repo

[Inflight services app wins 2017](#)

Fight4Flight

2017 · cathay-winner · runner-up · copy-pitch · high

Delay notification and re-planning from the passenger side.

Strength. Closest ancestor of Reconnect

Weakness. No transfer-desk owner

Takeaway. Passenger-only re-plan loses. The desk is the missing 2017 piece.

No public repo

[2017 top six ideas](#)

Aviator

2017 · cathay-winner · runner-up · foil · high

Dining and activity manager aimed at connecting passengers.

Strength. Saw connections as a moment

Weakness. Lifestyle layer on top of misconnect, not recovery

Takeaway. Connecting passengers are the moment; recovery is the job.

No public repo

[2017 top six ideas](#)

GME / Asia Miles visualiser

2021 · cathay-winner · champion · copy-pitch · medium

Map and proximity for Asia Miles partners, plus redemption notices.

Strength. Named loyalty BU

Weakness. Lifestyle, not ops

Takeaway. A visual of a real Cathay ledger (queue, slack) beats a generic dashboard.

[GitHub](#)

[markmcwong/cathay-hackathon-2021](#)

Cathay Green

2022 · cathay-winner · champion · foil · high

Currency and AI to promote carbon offset and green practice.

Strength. ESG as a named Cathay theme that year

Weakness. Easy to fake ESG in 2026 feasibility judging

Takeaway. Only claim sustainability if we can count fewer stranded hotel nights. Do not fake it.

No public repo

[5th Annual Cathay Hackathon](#)

NAAR cargo dimensioning

2023 · cathay-winner · champion · copy-pitch · high

Computer vision to size cargo and pack denser, with less manual labour.

Strength. Cargo \$ and labour

Weakness. Hardware/CV, not passenger recovery

Takeaway. Lead with a number (minutes, pax, queue length). NAAR led with accuracy and labour.

No public repo

[6th annual Cathay Hackathon](#)

GingTrip modular trolley

2024 · cathay-winner · champion · copy-pitch · high

Modular galley trolley that sorts cabin waste and feeds a dashboard.

Strength. Touchable theater

Weakness. Physical build is not our path

Takeaway. Do not add a robot. Do make the typhoon click as tactile as a trolley.

No public repo

[Aircraft Interiors: GingTrip 2024](#)

U-Explore (Kairos Takeoff Crew)

2025 · cathay-winner · champion · copy-pitch · high

All-in-one HK Express customer-journey platform: loyalty, community, gamified content.

Strength. Lived traveller pain

Weakness. Lifestyle; weaker under 2026 feasibility scoring unless a BU owns a 30-day pilot

Takeaway. Do not copy the loyalty app. Copy the named BU and the lived pain sentence.

No public repo

Swire: 8th annual Cathay Hackathon

Niki Jr.

2025 · cathay-winner · runner-up · foil · high

Polaroid robot named after CX DC-3 Niki; app booking, crowd-aware queueing.

Strength. Physical theater

Weakness. Not an ops problem

Takeaway. Presentation award is theater. We already have buttons; narrate them.

No public repo

HKUST ISD: Niki Jr.

FlyLab pre-flight briefings

2025 · cathay-winner · champion · copy-engineering · high

AI summaries of existing pre-flight briefing packs for pilots.

Strength. Real CX roles on the team

Weakness. Flight deck, not transfer desk

Takeaway. Closest 2025 analogue. Pair with a transfer agent. Do not invent a new PSS.

No public repo

Swire: 8th annual Cathay Hackathon

Anita chatbot

2018 · cathay-winner · runner-up · foil · medium

Chatbot that reminds passengers before boarding.

Strength. Proactive notify

Weakness. 2018 overall champion name not found in primary press

Takeaway. Notify is not recover. We notify after an agent approves a protected flight.

No public repo

LinkedIn: Anita first runner-up 2018

Lugless

2018 · cathay-winner · runner-up · foil · high

Bags packed and waiting at destination.

Strength. Named a physical pain

Weakness. Logistics company, not a desk tool

Takeaway. Bags are out of SCOPE. Do not add them before Saturday.

No public repo

[LinkedIn: Lugless second runner-up](#)

Cathay-Cargo box sequencing

2023 · cathay-repo · participant · foil · high

Python to sequence boxes; CV sizing; zip of UI. NAAR-adjacent, not the champion repo.

Strength. Cargo packing is a real labour problem

Weakness. 24h prototype, 1 star

Takeaway. Participant GitHub is not how NAAR won. Do not chase stars.

[GitHub](#)

[gd-Sahat/Cathay-Cargo](#)

CathayConnect

2023 · cathay-repo · participant · foil · high

Azure OpenAI trip planner. README includes the full pitch script.

Strength. Pitch script is a useful foil

Weakness. Generic super-app

Takeaway. Steal the habit of a written pitch. Do not steal the ChatGPT planner.

[GitHub](#)

[angusf777/Cathay-Hackathon-2023](#)

ReservaCathay

2023 · cathay-repo · participant · foil · high

ML overbooking plus GPT-personalised compensation offers.

Strength. Named overbooking as an airline problem

Weakness. LLM decides compensation — opposite of our invariant

Takeaway. Use this as the foil when a judge asks why we did not put GPT on ranking.

[GitHub](#)

[rrickyp/ReservaCathay](#)

Flymate

2023 · cathay-repo · participant · copy-pitch · high

Flutter app for travellers with disabilities: documents, stays, needs.

Strength. PRM is a real HKG transfer constraint

Weakness. Passenger app, not a desk queue

Takeaway. Wheelchair +15 min is already in mct.ts. Say it out loud.

[GitHub](#)

[NickMC05/CathayHackathon2023-Group44](#)

Syd (ctrl+shift+elite)

2023 · cathay-repo · participant · foil · high

React + OpenAI + DialogFlow travel buddy chatbot.

Strength. Typical 2023 student stack

Weakness. API keys in README, no ops invariant

Takeaway. This is what 2026 BU judges are tired of.

[GitHub](#)

[BrokenSpaghetti02/cathay-hack-T200](#)

TravelSync

2023 · cathay-repo · participant · foil · high

Vue app: chat, hotels, airport transfer, Splitwise.

Strength. Packaged a trip

Weakness. Super-app, not a Cathay desk

Takeaway. Do not add hotels to the Saturday demo.

[GitHub](#)

[TravelSync_Cathay](#)

PassportScanner

2018 · cathay-repo · participant · copy-pitch · high

Scan passport to auto-fill online check-in. Pain: <30% online check-in.

Strength. Started from a stated conversion number

Weakness. Stores passport images in the cloud — a 2026 non-starter

Takeaway. A number in the pitch is good. PNR in LLM prompts is already constrained in the pilot note.

[GitHub](#)

[alifarooqi/PassportScanner](#)

Carbon per passenger model

2022 · cathay-repo · participant · foil · medium

ML for carbon emissions per passenger per flight.

Strength. Matches 2022 theme

Weakness. Not a desk workflow

Takeaway. Theme-chasing without a desk owner is 2022, not 2026.

[GitHub](#)

[Cathay-Hackathon-2022](#)

Optimal flight route (2025 repo)

2025 · cathay-repo · participant · foil · medium

Public 2025 repo: most optimal flight route. Not the student champion.

Strength. Network problem is real

Weakness. Champion U-Explore has no public repo; this is the GitHub you will find instead

Takeaway. Search GitHub for 2025 and you will not find Kairos. That is the point.

[GitHub](#)

[anoushkapant/CathayHackathon2025](#)

Amadeus Passenger Recovery

2018 · production · production · incumbent · high

Mass reaccom: >50 flights and 8,000 pax in 40 minutes. CX live since ~2018 with Accenture chatbots and 15below.

Strength. Cathay already paid for this

Weakness. Does not walk UM/WCH through T1 in the last 90 minutes

Takeaway. Pitch beside this tool. Never instead.

No public repo

[Amadeus Passenger Recovery · Typhoon Trouble \(Airport Industry Review, 2019\)](#)

CX typhoon programme (Rogers 2019)

2019 · production · production · copy-pitch · high

Josh Rogers: 100,000 customers when the hub closes; wanted airborne reprotect on inflight Wi-Fi.

Strength. Named HKG typhoon as a yearly business challenge

Weakness. That airborne-misconnect layer is still the gap

Takeaway. Our CX254 +180 demo is the 2019 wish, on a desk iPad.

No public repo

[Typhoon Trouble](#)

United ConnectionSaver

2019 · production · production · copy-pitch · high

Holds the outbound using walk time; SMS with minutes. 3.3 million connections saved since 2019.

Strength. Walk time is a product

Weakness. Hold vs rebook is a different decision; out of our SCOPE

Takeaway. Encode walk + WCH + UM now. Hold-the-flight is LATER. Know the map.

No public repo

[United ConnectionSaver launch](#)

NASA Open MCT

2015 · aerospace · production · copy-engineering · high

Web mission-control glass: telemetry, timeline, mobile, one place.

Strength. One glass, no tab-hopping

Weakness. Spacecraft telemetry, not PNRs

Takeaway. Queue + connection + action drawer is already one glass. Say mission control, not dashboard.

[GitHub](#)

[nasa/openmct](#)

SpaceX flight software culture

2012 · aerospace · production · copy-engineering · high

Control cycle on a clock; config DSL for non-software engineers; sim + HITL; no LLM in the loop.

Strength. Determinism

Weakness. Not open source; culture analogue only

Takeaway. packages/engine is already this shape. Pitch it that way.

No public repo

[Stack Overflow: software that flies Starships](#)

amadeus-disruption-agent

2026 · irops-hack · research · copy-engineering · high

LangGraph + Amadeus APIs. LLM phrases verified facts only. RAG for EU261.

Strength. Same invariant as Reconnect: model does not decide

Weakness. Passenger-side, not HKG desk

Takeaway. Cite reasoning[] the way they cite policy passages.

[GitHub](#)

[amadeus-disruption-agent](#)

AutoRescue

2025 · irops-hack · participant · foil · high

AWS Bedrock + Amadeus search for cancelled-flight rebooking.

Strength. Live Amadeus, not only mocks

Weakness. LLM-orchestrated decisions

Takeaway. Live Amadeus is impressive.
Decision authority is not transferable to Saturday.

[GitHub](#)

[AnuR-lab/AutoRescue](#) · [Devpost: AutoRescue](#)

HKG A-CDM

2010 · production · production · copy-engineering · high

TOBT/TSAT shared clock for turnaround.
Missing passenger is an A-CDM delay reason.

Strength. Airport clock of record

Weakness. Turnaround, not PNR recovery

Takeaway. Never use the iPad wall clock. FIDS
eventTime / sim clock only.

No public repo

[HKIA A-CDM Operations Guidelines](#)

Reconnect (this repo)

2026 · cathay-repo · participant · copy-engineering · high

Deterministic MCT engine + iPad triage +
guarded draft. Pilot: Altéa read, FIDS MQTT,
CANS push.

Strength. Zero LLM in engine

Weakness. No quantified impact yet

Takeaway. The product is already the 2026
feasibility shape. The pitch must not throw it
away.

No public repo

[docs/PILOT_PROPOSAL.md](#) ·
[packages/engine/src/mct.ts](#)

Accenture Rebooking Experience

2018 · production · production · incumbent ·
high

CX self-rebook chatbots in the typhoon stack:
notify, then four steps to pick a preferred flight.

Strength. Passenger self-help hours ahead

Weakness. Not a transfer-desk queue; does not
escort UM/WCH through T1

Takeaway. Chatbot self-rebook is already
bought. Our send is agent-approved CANS, not a
second bot.

No public repo

[Typhoon Trouble \(Airport Industry Review, 2019\)](#)

15below customer communication

2018 · production · production · incumbent · high

Notification layer on CX disruption: once a new journey is accepted, 15below sends the new flight details.

Strength. CX already paid for proactive passenger comms

Weakness. Mass notify, not last-90-minute special handling

Takeaway. CANS after human confirm sits beside 15below, not over it. Do not pitch a new notification platform.

No public repo

[Typhoon Trouble \(Airport Industry Review, 2019\)](#)

How to use this pack

Serve the folder (Chrome will not fetch JSON from `file://`):

```
python3 -m http.server 4173 --directory docs/briefing
```

Then open `http://localhost:4173`. Print to PDF from the browser, or use the generated `Reconnect-competitive-intel.pdf`. Spreadsheet: `data/projects.csv`. Markdown: `docs/competitive/`. Present: add `#slides` and use `← →`.